

BNSAT-R

BITCOIN NODE SECURITY, AVAILABILITY & RESILIENCE RESEARCH

Programme Concept Note

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Geographic Base: Kampala, Uganda

Initial Research Project: BNOCR - Bitcoin Node Operational Continuity
Research

Research. Measure. Build. Contribute.

1. Executive Summary

UBRO CLASSIFIEDS LIMITED proposes to establish BNSAT-R - Bitcoin Node Security, Availability & Resilience Research - as an independent research programme focused on the operational security and resilience of Bitcoin infrastructure.

BNSAT-R is intended to contribute to the wider Bitcoin ecosystem through independent research, controlled experimentation, open-source software, technical documentation and collaboration with experienced Bitcoin developers and researchers.

The programme's initial research project is BNOCR - Bitcoin Node Operational Continuity Research. BNOCR will investigate the operational recovery and continuity of independently operated Bitcoin full nodes following controlled disruptions to power, network connectivity and system resources.

The central research question is: How do independently operated Bitcoin full nodes recover and return to operational continuity following controlled disruptions, and can this recovery process be measured using a reproducible framework?

The research is deliberately positioned within an existing ecosystem of Bitcoin protocol development, node research, infrastructure studies and open-source engineering. UBRO does not claim that the proposed research area is unexplored. Instead, the first phase of BNSAT-R will validate the research gap through systematic review of existing work and consultation with experienced Bitcoin researchers.

The programme will adopt five principles: research before building; measure before claiming; collaborate before duplicating; reproduce before generalising; and contribute openly where appropriate.

2. About UBRO CLASSIFIEDS LIMITED

UBRO CLASSIFIEDS LIMITED is a Uganda-based IT solutions provider with experience in information technology infrastructure, systems administration, networking, software development, systems integration, server management, database administration, IT operations and technical training.

UBRO has practical experience working with technology infrastructure and organisations operating in environments where cost, availability, connectivity and technical capacity are important operational considerations.

The organisation is now seeking to extend this experience into independent open-source research concerning Bitcoin infrastructure. BNSAT-R represents a long-term commitment to developing technical research capacity rather than a one-off project.

3. Programme Vision

The vision of BNSAT-R is:

To contribute independent, reproducible and open-source research that improves understanding of the operational security, availability and resilience of Bitcoin infrastructure.

The programme seeks to contribute to Bitcoin's decentralisation by expanding the diversity of people and organisations capable of conducting meaningful security and infrastructure research.

4. Programme Mission

The mission of BNSAT-R is to conduct independent Bitcoin infrastructure research; develop reproducible experimental methodologies; study operational resilience; develop useful open-source research tools; build technical research capacity in Uganda; collaborate with the international Bitcoin

development community; and publish research findings openly where appropriate.

5. Research Pillars

Pillar 1 - Node Recovery: Understanding how Bitcoin full nodes recover following operational disruptions.

Pillar 2 - Node Availability: Investigating how infrastructure conditions affect node availability and continuity.

Pillar 3 - Resource Efficiency: Studying how resource-constrained environments affect node operation and recovery.

Pillar 4 - Operational Security: Investigating practical methods for maintaining secure and reliable Bitcoin infrastructure.

6. Initial Research Project

The first BNSAT-R research project will be BNOCR - Bitcoin Node Operational Continuity Research.

The project will investigate how Bitcoin full nodes behave during and after controlled operational disruptions.

The initial project is not intended to test Bitcoin's consensus rules. It will not attempt to modify Bitcoin Core. It will not interfere with public Bitcoin infrastructure. Instead, it will examine the operational layer surrounding individual node deployments.

7. Central Research Question

How do independently operated Bitcoin full nodes recover and return to operational continuity following controlled disruptions, and can this recovery process be measured using a reproducible framework?

8. Secondary Research Questions

1. How does the type of disruption influence node recovery?
2. What factors delay recovery?
3. How do hardware and resource profiles influence recovery?
4. How does network connectivity affect recovery?
5. What is the relationship between node restart and full operational recovery?
6. How much manual intervention is required?
7. Can node recovery be described using standardised operational states?
8. Can a reproducible benchmark methodology be developed?

9. The Proposed Research Gap

Existing Bitcoin research has addressed many areas, including consensus security, cryptographic security, peer-to-peer networking, node performance, resource requirements, network decentralisation and infrastructure resilience.

BNSAT-R proposes to investigate whether there remains a useful research gap concerning the systematic measurement of the operational recovery pathway of individual full-node deployments following controlled disruptions.

This is currently treated as a research hypothesis rather than an established fact. The first phase of BNOCR will therefore include a formal review of academic research, Bitcoin Core research, Bitcoin

developer discussions, existing node monitoring tools, existing benchmarking tools, distributed-systems research and infrastructure resilience research.

The research design will be adjusted based on the findings.

10. Proposed Recovery Model

R0 - Failure: The node is unavailable.

R1 - Process Recovery: Bitcoin Core process successfully starts.

R2 - System Recovery: The underlying operating environment and required services are functioning.

R3 - Peer Recovery: The node establishes normal peer connectivity.

R4 - Chain-State Recovery: The node reaches the expected blockchain state. R5

- Operational Recovery: The node resumes normal operation.

R6 - Verified Continuity: The node remains stable for a defined observation period.

The final model will be validated during the methodology development stage.

11. Research Variables

Potential controlled disruptions may include power interruption, unexpected shutdown, network interruption, intermittent connectivity, bandwidth limitation, CPU constraints, memory constraints and storage pressure.

The final experimental variables will be selected after the literature and prior-work review.

12. Research Methodology

Phase 1 - Prior Work Review: Identify existing research and tools.

Phase 2 - Research Gap Validation: Determine whether BNOCR addresses a meaningful gap.

Phase 3 - Experimental Design: Develop a reproducible methodology.

Phase 4 - Controlled Testing: Conduct experiments in an isolated laboratory.

Phase 5 - Replication: Repeat experiments to assess consistency.

Phase 6 - Analysis: Evaluate results statistically and qualitatively.

Phase 7 - Publication: Publish findings and appropriate open-source tools.

13. Uganda as the Research Base

BNSAT-R will initially operate from Kampala, Uganda.

The programme will not assume that Uganda represents all Bitcoin users or all infrastructure environments. Instead, Uganda provides the research team with a practical environment from which to investigate operational questions concerning electricity availability, Internet connectivity, bandwidth, hardware affordability and technical skills.

Future research may expand to other Ugandan locations, potentially including Jinja, Mbarara and Mbale. Any geographic expansion will be based on evidence and research requirements.

14. Open-Source Commitment

BNSAT-R will adopt an open-source-first approach.

Where appropriate, UBRO intends to release experimental scripts, monitoring tools, data processing tools, benchmarking tools, methodology documentation and non-sensitive research datasets.

The project will seek to contribute improvements to existing open-source projects where that is more appropriate than creating new software.

15. Security and Ethics

All experiments will be conducted in controlled environments.

The research will not attack public Bitcoin nodes, conduct unauthorised penetration testing, disrupt public networks, perform denial-of-service attacks or probe third-party infrastructure without permission.

The research laboratory will remain isolated from production systems.

16. Long-Term Research Programme

Following the initial research, BNSAT-R may expand into Bitcoin infrastructure resilience under diverse operational conditions; resource-constrained Bitcoin infrastructure research; Bitcoin infrastructure resilience in multiple geographic environments; and research into future Bitcoin security challenges, including potential quantum-readiness questions.

Future research topics will be selected based on evidence and community priorities.

17. Expected Contribution

UBRO seeks to contribute independent research, open-source tools, reproducible methodologies, technical documentation and African research capacity, while building new contributors to Bitcoin security research.

The programme's long-term objective is to demonstrate that meaningful Bitcoin security research can be conducted by technically capable organisations outside traditional Bitcoin development centres.

18. Conclusion

BNSAT-R represents UBRO CLASSIFIEDS LIMITED's proposed entry into independent Bitcoin security and infrastructure research.

The programme begins with a deliberately focused research question concerning the operational continuity and recovery of Bitcoin full nodes.

UBRO recognises the extensive body of existing Bitcoin research and does not seek to duplicate work already completed. The programme's first responsibility is therefore to listen, research, validate and collaborate.

UBRO seeks technical feedback from experienced Bitcoin developers and researchers before committing to a full research programme.

The organisation's long-term goal is to make meaningful, open-source contributions to the Bitcoin ecosystem while building sustainable technical research capacity in Uganda.

Status and Request for Ecosystem Feedback

Status: Research Concept - Seeking Technical Review and Ecosystem Feedback

This document represents an initial research concept and is not presented as a claim of established novelty. UBRO CLASSIFIEDS LIMITED welcomes feedback from Bitcoin developers, researchers and open-source contributors to refine the research direction and avoid duplication of existing work.

UBRO is particularly interested in technical guidance concerning the relevance of the research question, existing work that may overlap with the proposed direction, appropriate methodology and opportunities for constructive collaboration.